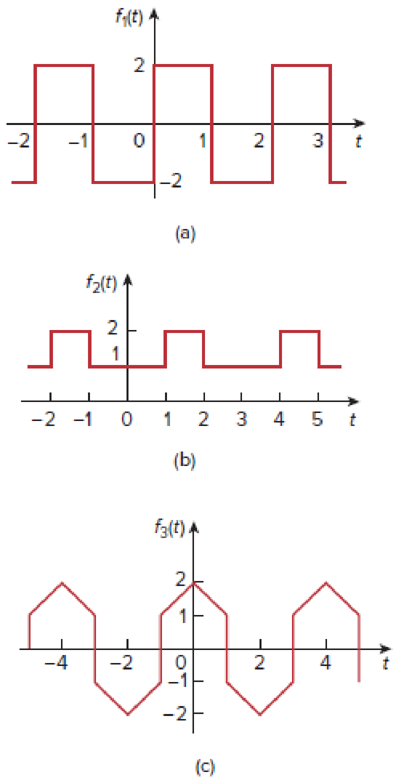
Chapter 17, Problem 18P

Problem

Determine the fundamental frequency and specify the type of symmetry present in the functions in Fig. 17.56.

Figure 17.56:



Step-by-step solution

Step 1 of 6

(a)

Refer to Figure 17.56 (a) in the textbook.

Consider the time period is $T = 2$.

Determine the fundamental frequency ω_o.

$$\omega_o = \frac{2\pi}{T} \quad \text{...... (1)}$$

Replace 2 for T in equation 1.

$$\begin{aligned} \omega_o &= \frac{2\pi}{T} \\ &= \frac{2\pi}{2} \\ &= \pi \text{ rad/sec} \end{aligned}$$

Therefore, the fundamental frequency is $\boxed{\pi \text{ rad/sec}}$.

Step 2 of 6

Consider the function $f_1(t)$ as,

$$f_1(t)=\begin{cases}-2 & -1<t<0 \\ 2 & 0<t<1\end{cases} \quad \text{...... (2)}$$

Substitute $t=-t$ in $f_1(t)$.

$$f_1(-t)=\begin{cases}-2 & 1>t>0 \\ 2 & 0>t>-1\end{cases} \quad \text{...... (3)}$$

From equation 2 and equation 3,

$$f_1(t)=-f_1(-t)$$

Therefore, the function $f_1(t)$ is $\boxed{\text{odd symmetry}}$.

Step 3 of 6

(b)

Refer to Figure 17.56 (b) in the textbook.

From Figure, the time period is $T=3$.

Determine the fundamental frequency ω_o.

Replace 3 for T in equation 1.

$$\begin{aligned}\omega_o &= \frac{2\pi}{T} \\ &= \frac{2\pi}{3} \\ &= \frac{2\pi}{3}\ \text{rad/sec}\end{aligned}$$

Therefore, the fundamental frequency is $\boxed{\frac{2\pi}{3}\ \text{rad/sec}}$.

Step 4 of 6

Consider the function $f_1(t)$ as,

$$f_2(t)=\begin{cases}2 & -1.5<t<1 \\ 1 & -1<t<1 \\ 2 & 1<t<1.5\end{cases} \quad \text{...... (4)}$$

Substitute $t=-t$ in $f_2(t)$.

$$f_2(-t)=\begin{cases}2 & 1.5>t>1 \\ 1 & 1>t>-1 \\ 2 & -1>t>-1.5\end{cases} \quad \text{...... (5)}$$

From equation 4 and equation 5,

$$f_2(t)=f_2(-t)$$

Therefore, the function $f_2(t)$ is $\boxed{\text{even symmetry}}$.

Step 5 of 6

(c)

Refer to Figure 17.56 (c) in the textbook.

From Figure, the time period is $T = 4$.

Determine the fundamental frequency ω_o.

Replace 4 for T in equation 1.

$$\begin{aligned}\omega_o &= \frac{2\pi}{T} \\ &= \frac{2\pi}{4} \\ &= \frac{2\pi}{4}\ \text{rad/sec}\end{aligned}$$

Therefore, the fundamental frequency is $\boxed{\frac{2\pi}{4}\ \text{rad/sec}}$.

Step 6 of 6

Consider the function $f_1(t)$ as,

$$f_3(t) = \begin{cases} t & -2 < t < -1 \\ t+2 & -1 < t < 0 \\ 2-t & 0 < t < 1 \\ t & 1 < t < 2 \end{cases} \quad \text{...... (4)}$$

Therefore, the function $f_3(t)$ is $\boxed{\text{halfwave even symmetry}}$.